

## Solution Requirements

|                      |                                                              |
|----------------------|--------------------------------------------------------------|
| <b>Date</b>          | 27 June 2026                                                 |
| <b>Team ID</b>       |                                                              |
| <b>Project Name</b>  | OptiCrop: Smart Agricultural Production Optimization Engine. |
| <b>Maximum Marks</b> | 4 Marks                                                      |

### Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviours (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

### Step 1: Functional Requirements (FR)

| S.No | Requirement Category            | Requirement Description                                                                                                                                     | Priority |
|------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1    | Soil & Environmental Data Input | The system allows users to enter Nitrogen (N), Phosphorous (P), Potassium (K), temperature, humidity, pH, and rainfall values required for crop prediction. | High     |
| 2    | Input Validation                | The system validates all entered values to ensure complete and valid data before processing the prediction.                                                 | High     |

|   |                             |                                                                                                                                                       |        |
|---|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 3 | Machine Learning Prediction | The system analyzes the input data using the trained machine learning model and predicts the most suitable crop.                                      | High   |
| 4 | Crop Suitability Analysis   | The system evaluates whether the selected soil and environmental conditions are suitable for the recommended crop.                                    | High   |
| 5 | Recommendation Dashboard    | The application displays the recommended crop along with prediction results in a clear and user-friendly interface.                                   | High   |
| 6 | Business Rules              | The recommendation is generated only after all mandatory input parameters are provided and fall within valid ranges supported by the trained model.   | High   |
| 7 | Agricultural Insights       | The system provides crop-related insights to help users understand the recommendation and support better farming decisions.                           | Medium |
| 8 | Future Enhancements         | The application is designed to support future features such as fertilizer recommendation, real-time weather integration, and crop disease prediction. | Low    |

## Step 2: Non-Functional Requirements (NFR)

| S.No | NFR Category               | Requirement Description                                                                                                          | Target Metric / Acceptance Criteria             |
|------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| 1    | Performance & Speed        | The application should process user input and display the crop recommendation without noticeable delay.                          | Prediction displayed within 2 seconds           |
| 2    | Scalability                | The system should support additional crop datasets, new features, and increased usage without major changes to the application.  | Supports future feature and dataset expansion   |
| 3    | Security & Data Privacy    | User inputs should be validated, and the system should process data securely without storing personal information unnecessarily. | 100% input validation with secure data handling |
| 4    | Reliability & Availability | The application should consistently generate accurate crop recommendations and remain stable during normal operation.            | 99% successful prediction rate during testing   |
| 5    | Usability & Accessibility  | The interface should be simple, intuitive, and easy to use for farmers with basic computer knowledge.                            | Users can complete prediction in 3 simple steps |

|   |                                  |                                                                                                                   |                                                 |
|---|----------------------------------|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| 6 | Maintainability<br>& Portability | The project should have a modular code structure and run on Windows, Linux, and macOS with minimal configuration. | Cross-platform support with modular Python code |
|---|----------------------------------|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|